

ARTICLE PUBLISHED IN 10 JUN 2025

ARTICLE PUBLISHED IN 21 NOV 2025

ARTICLE PUBLISHED IN 20 NOV 2025

From Assistant to Autonomous Partner

(my first contact with Google's Antigravity)

For too long, AI coding tools felt like an autocomplete on steroids. They were reactive, waiting for a prompt, and only capable of tackling discrete, simple tasks. I thought things would stay this way for longer, but Google has released its new Antigravity, an advanced agentic development platform designed to enhance coding projects.

Antigravity embraces this philosophy, moving us from "AI-assisted" to "agent-native" development. At its core lies Gemini 3 Pro (and notably, supports other models like Claude Sonnet 4.5 and GPT-OSS) to not just suggest the next line of code, but to take true, autonomous action across the three core development surfaces: the code editor, the terminal, and the browser.

I genuinely believe the future lies in agentic AI—entities that can reason, plan, and execute complex workflows on their own. So I was very excited and decided to give Antigravity a shot and see what it was all about.

A day flowing in antigravity!

The first step was finding a case, a project that I could take using it. I've been working on deploying a prediction model for a client lately. Although this is a new challenge, with its own characteristics and unique architecture, it also is the type of project I have done many many times over, thus using antigravity to revisit old designs and models seemed a perfect opportunity.

The very first thing that has impressed me is how serious it takes the critical challenges of agentic systems: transparency, accountability, and seamless integration. My first takeaway is that Antigravity isn't just a new feature for an IDE—it's the architecture for the "AI-infused" enterprise I've been discussing for months.

My interactions were pretty much like if I was supervising a junior developer, literally "debating" and explaining how I wanted the model to be developed, how data should be treated, how functions should be called, how tests should be performed and how the whole thing should be integrated and deployed.

Though the experience, four things stood out:

1. The Artifacts System: Trust Through Verifiable Outputs

The "black box" is the silent killer of trust in AI. How can you sign off on code if you can't verify the steps the agent took? Antigravity solves this with Artifacts: structured, verifiable evidence of the agent's actions.

I noticed that instead of sifting through thousands of raw logs, I was presented with clear Task Lists, Code Diffs, Screenshots of the UI after changes, and even Browser Recordings of end-to-end tests. It had an interesting impact on me, as I moved from line-by-line debugging to high-level review and architecture, elevating my analysis and judgment by automating the tedious execution and providing audit-ready proof.

It felt like Antigravity has accountability built in by design.

2. Cross-Surface Agentic Control

An agent is only as good as its access to the environment. Antigravity agents are truly cross-surface, meaning they can perform things like Write and Refactor Code in the editor, Run Terminal Commands to install dependencies, run tests, and spin up a local server.

One cool feature that is how it acts on the browser to test their own work in a real Chrome instance, verify the output, and even record the user flow.

I delegated a task—"Build a small Python app with a dynamic event trigger and full search functionality"—and the agent autonomously handled the dependencies, the coding, the local launch, and the test validation. It delivered a ready-to-review result and its own proof of work. This is no small potatoes!

3. Multi-Agent Orchestration with Manager View:

For large-scale enterprise projects, we know that successful architecture is about orchestration, not isolated tools. Antigravity has the Manager View, a "mission control" type of dashboard designed for high-level oversight.

I am impressed with parallelism: I was able to spawn and monitor multiple agents—a Refactoring Agent cleaning up an authentication module in one workspace while another was implementing a new front-end feature in a parallel workspace. This is the practical realization of multi-agent systems in production, moving the developer's role to that of an architect and orchestrator, maximizing velocity by parallelizing tasks that were previously sequential and context-switching heavy.

4. Continuous Feedback Loops and Self-Improvement

Integrated Feedback that kept context and persistence: Antigravity allows you to add Google Docs-style comments directly onto any Artifact—a screenshot, a plan, or a code diff—without stopping the agent's work.

The agent seamlessly incorporates this feedback and continues its execution, ensuring continuous refinement. This solves the "20% problem" I've seen derail so many POCs.

Although it is not full knowledge management capability, it sure is a strong step towards it. Far greater than what I had seen. Agents can learn from past projects, storing successful code snippets, execution patterns, and architectural decisions to an internal knowledge base. This means your agents get smarter, and your enterprise memory grows with every project.

After the Rush

As I am writing this, I was able to achieve in one day what used to take me a full sprint of work, full of syncs, dailies, code-review sessions, refactoring and who-knows-how-many versions. This has strengthened my belief that this type of AI development, embodied by Antigravity, is the future.

It's an urgent call for us as leaders—CTOs, CDOs, and Product Heads—to step up and lead the charge.

Antigravity and its competitors will redefine Roles, STARTING YESTERDAY. However we need to view this not as a job replacement, but a skill elevation. Our junior developers become "AI Supervisors," learning to architect and validate rather than just implement line-by-line.

It is not all praises and high sugar rush though. Documentation of the final artifact result is the low-point that I believe deserves attention from Google's development team. I might be wrong but I think there is still a lot to do till Antigravity can autonomously run a full Dev/Sec/Ops cycle on its own. I might have missed something but if you are running a tech team not only focused on developing brand new things but also sustaining, patching, and updating legacy code and modes, the documentation on how things were built, the vulnerability analysis, the governance structure, the quality test results and other aspects central to a decent technology area will still require a serious dose of old-fashion human work.

This is the perfect segway for an old message of mine, still fresh and relevant: Focus on Governance and Observability! The complexity of autonomous agents demands a robust central nervous system. We must build the ethical and operational guardrails to ensure agents are effective, compliant, and accountable. People is still central, documentation is even more important and governance is paramount!

With that caveat, Antigravity is promising and easy to use. I believe it is capable of delivering innovation at a velocity we could only dream of before.

Let's keep this vital conversation going.

#AI #AgenticAI #DigitalTransformation #GoogleAntigravity #Gemini3 #FutureOfWork #CTO #CDO #Innovation #DataEngineering #SoftwareDevelopment

ARTICLE PUBLISHED IN 18 SEPT 2025

My Reflections on the AI Agent Playbook

If you're in the trenches of data science and AI, you know the conversations are shifting. It's no longer just about building cool models or automating a single task. We're talking about AI agents—entities that can reason, plan, and execute complex workflows on their own.

But here's the cold, simple truth I learned in practice this summer: the most successful AI initiatives are not about trying to rip out entire infrastructures, nor about building AI expert teams isolated from IT or big data competencies and starting from scratch. That's a fool's errand. The real competitive advantage is being unlocked by a far more pragmatic strategy: matching AI agents to the existing human processes and workflows already in place.

This isn't a sign of limited vision. It's a two-step master plan for success.

First, you integrate. You find a high-value process—maybe a customer quote system or a KYC analysis workflow—and you deploy an agent to make it better. You get a quick win, a measurable return on investment (ROI), and you start building trust and momentum within the organization. This is about being smart and strategic, not about making a big, noisy announcement. It's about building a foundation of data and buy-in for what comes next.

Then, you transform. Once you've proven the value, you can begin the more ambitious, transformative work. You start to fundamentally redesign your core workflows around the unique capabilities of AI. This isn't just about working faster; it's about reimagining how work gets done. Think about a product development cycle that used to take two years, now being completed in four months because you've parallelized sequential human activities with AI. That's not a marginal improvement—that's an exponential leap.

This phased approach is what separates the companies that get a little bit better from those that become "AI-infused".

This strategy however falls victim to its own success as the explosion of specialized AI agents—from a data agent in Snowflake to a conversational one in AWS—has created a new kind of architectural problem. Your agents can't talk to each other. They're siloed, and they can't collaborate on complex tasks.

This is why interoperable AI agent frameworks are no longer a luxury—they are a strategic imperative.

These frameworks act as the central nervous system for your AI ecosystem. They provide the protocols and standards that allow different agents, models, and tools to communicate and collaborate seamlessly. Without this, your multi-agent system is just a collection of disconnected tools.

We're already seeing two key protocols emerge to solve this. The Model Context Protocol (MCP) standardizes how AI systems connect to various datasets and APIs. This means your developers aren't writing custom code for every single integration, which saves time and money. It's the plumbing that ensures your agents can pull data from a dozen different sources without a hitch.

The other piece of this puzzle is the Agent Communication Protocol (ACP). This is the common language that allows agents themselves to talk to one another. It enables them to negotiate tasks, resolve conflicts, and work together as a cohesive unit. This is critical for complex operations, like a supply chain where multiple agents are managing warehousing, shipping, and inventory in real-time.

Organizations that get locked into a single-vendor ecosystem risk being outpaced by more agile competitors. The ability to mix and match specialized agents from different providers gives you the flexibility to adapt to changing market demands and stay ahead of the curve.

This central nervous system for AI is my key learning point of the summer, and I am proud of the success I achieved over it. It accounts for AI effectiveness, compliance and coordination.

Companies do not invest in AI because it makes them look cool (though some do). They invest because AI has to be effective: cutting costs, augmenting productivity, providing better experiences. This means this central nervous system has to measure and proof beyond doubt that the AI agents are effective, productive and impactful.

They cannot do so at any cost, nor by acting rogue. There are rules and regulations to be followed, not only the most obvious ones like GDPR, EU AI Act, Digital Services Act but also company policies, authorization paths, log registries, shipping regulations, therefore making sure that each and every AI agents must stay compliant and their assembled work must also be compliant to rules.

Finally coordination is needed to make sure that those AI agents that have been developed by different teams, for a variety of purposes can evolve and grow without stepping on each other's toes as they stay active for long.

I have talked about this many times, it has been true for a while and will continue to be true in the future: This central nervous system must also rest on three non-negotiable pillars into the AI lifecycle: observability, accountability, and governance.

Observability: Think of this as the eyes and ears of your AI system. It's about continuous monitoring of your data, models, and infrastructure. Without it, your models can "silently degrade over time" due to changes in data or a shift in the problem you're trying to solve. You won't know there's a problem until it's too late. Observability provides the visibility to detect issues before they impact your bottom line.

Accountability & Governance: These are the ethical and operational frameworks that keep your AI systems in check. You need policies and procedures to ensure ethical deployment, manage risk, and protect stakeholders. This includes things like:

Fairness and Bias Mitigation: You must ensure your AI agents don't discriminate. This means using diverse datasets and regularly auditing your models for bias.

Transparency and Explainability: The inner workings of your AI systems need to be open to scrutiny. We need to be able to explain why a model made a particular decision, especially in high-stakes fields like healthcare or finance.

Accountability: Humans, not machines, must take ultimate responsibility for AI-driven decisions. Your governance framework must provide an audit trail that ties model behavior to specific owners.

The most successful AI strategies are not those that attempt a radical, disruptive overhaul. They are defined by a phased, human-centric approach that begins with integration and evolves into strategic transformation. This journey validates a simple truth: matching AI agents to existing processes is the recipe for early success.

The future of work is not a zero-sum game between humans and machines. It is a symbiotic partnership. AI agents are the perfect partners for automating routine work and providing predictive insights. Human expertise remains crucial for creativity, empathy, and ethical judgment.

By investing in interoperable frameworks, establishing rigorous governance, and prioritizing continuous observability, we can ensure our AI systems are not just powerful, but also reliable, accountable, and aligned with human values. This is how we confidently navigate the intelligent age and forge a new blueprint for a human-machine partnership that will define the next era of business transformation.

My final summer message: AI is a different beast, it is NOT like software development. Run the same software a million times over and it will perform exactly as it was coded every single time. Run the same prompt twice and you will get different results even so slightly. Agentic AI is an entity, not a program. It evolves, changes,

drifts, adapts, improves. AI is the unavoidable future, its adoption is necessary but the sooner CTOs and CEOs understand they are not dealing with lines of code, the better.

ARTICLE PUBLISHED IN 15 JUL 2025

From Input-Output to Intent: Architecting the Future of Enterprise AI

I want to thank you all for the feedback on the piece I published last week. I started writing that piece as a way to settle down my own ideas for live architecture and as I went on experimenting them on Google Cloud, other critical aspects of deployment kept surfacing, demanding tweaks in the architecture, addition of connectors, setup of verification points and KPIs that pushed my thoughts towards the most efficient way to deploy Agentic AI in a scalable fashion, an AI architecture framework for enterprises of sorts

We're no longer talking about AI as a tool that simply takes an input and gives you an output. The next generation of AI is fundamentally different. These systems are moving from being reactive to being proactive, from executing commands to actively reasoning, planning, and executing actions on their own. This is the emergence of agentic AI, and it's poised to transform how we approach intelligent automation. For enterprises this is the ultimate departmental integration.

And here lies a catch I've seen before: deploying agentic systems in an enterprise environment requires more than just adopting the latest LLM models or "vibe-coding" techniques. Success demands architectural patterns that balance cutting-edge capabilities with organizational realities as one change in the marketing process might have huge implications for legal, marketing or logistical departments. That's what this architecture aims to handle. It is not about IT but how agentic AI connects itself with the overall organization through the IT stack.

We're talking about a familiar set of topics I've written before: governance requirements, audit trails, security protocols, and, critically, ethical accountability. Amazing how the fundamentals do not change, they only apply to different situations!

Just as I've said time and time again, building a robust data governance framework is a journey, not a destination. The same holds true for architecting agentic AI. You can't simply flip a switch and expect a fully autonomous system to operate safely and effectively within your business.

In my experience, organizations that succeed in implementing data products share the same trait: they prioritize simple, composable architectures over overly complex frameworks. Allowing them to manage complexity while controlling costs and maintaining performance standards. The implementation must be measured, evaluated and constantly discussed with the teams, partners, internal stakeholders and users.

It is the classic data digital transformation approach, acting as the background constantly evaluating when predictability and control take precedence versus flexibility and autonomous decision-making. It is on this background that the architectural framework will hang,

This framework is a systematic maturity progression, designed to help organizations build competency and stakeholder trust incrementally before they ever attempt to advance to more sophisticated implementations. It anticipates emerging regulatory frameworks like the EU AI Act and others, making it a blueprint for future-proof development.

We start with The Foundation Tier which is the essential first step for any enterprise looking to deploy agentic AI, focusing on establishing trust and governance before attempting greater autonomy. It is defined by three key architectural patterns: tool orchestration, which restricts the AI to a curated set of approved APIs and data sources; reasoning transparency, which requires logging and auditing the AI's decision-making process to ensure explainability and accountability; and the disciplined use of data lifecycle patterns, which guarantees

that the data the AI consumes and generates is validated, well-managed, and ethically handled. This tier serves as the bedrock, proving the system's reliability and building the crucial buy-in from stakeholders by demonstrating a controlled and auditable approach to AI development. This is also where most companies and consulting companies are as of today.

The second tier focuses on Workflows. It builds upon the foundation of trust established in Tier one, delivering significant automation by orchestrating multi-step, predefined processes. This tier is not about full autonomy but about creating highly efficient, repeatable AI workflows. It is characterized by five core patterns: Prompt Chaining, which links a series of deterministic LLM calls; Routing, which directs tasks to the appropriate sub-process; Parallelization, which enables concurrent task execution; the Evaluator-Optimizer pattern, which introduces a self-correction loop for improving output quality; and the Orchestrator-Workers model, which delegates complex tasks to specialized agents. These features allow automation of complex business processes while maintaining a predictable, auditable framework with clear human oversight, measuring the relevant KPIs and validating (or not) efficiency and productivity gains.

The final tier is all about autonomy, and represents the advanced stage of agentic AI deployment, enabling systems with genuine, goal-directed planning capabilities. This tier is not about replacing human decision-making with fully autonomous systems, but rather about deploying Constrained Autonomy Zones that operate within a tightly controlled, well-governed environment. The core principle here is to allow the AI to dynamically determine its own approach to achieving a goal, while still embedding validation checkpoints and human oversight to maintain accountability and mitigate risk. This introduces flexibility and innovation, allowing the AI to learn and adapt, but always within boundaries that align with regulatory compliance and ethical standards.

Industry-Specific Implementation: A Tale of Context

One size does not fit all. The general framework I've outlined is a guide, but any specific implementation must be tailored to the unique guiding principles and utility functions of your industry: regulatory constraints, business objectives, tech stack limitations.

I know first hand that Financial Services, Healthcare, Gaming, Retail and Manufacturing have specificities that make implementation as different as day and night, so in the coming weeks my idea is to explore contextual engineering as a core feature of any AI deployment.

Finally I would like to share how lucky I am to have intellectual partners in this journey, like Subash Natarajan and Ahilan Ponnusamy whose work have inspired me and Naomi Garratt whose tech prowess always impresses me.

ARTICLE PUBLISHED IN 08 JUL 2025

Serving AI in Real-Time: Architecture for Low-Latency, High-Availability Inference

As I've said time and time again, in production environments, deploying a trained AI model is no longer a matter of saving a .pkl file and calling it a day. Real-time inference requires architectural discipline. This is especially true in latency-sensitive use cases like finance, healthcare, gaming, real-time bidding, or fraud detection, where every millisecond counts and system failures carry significant business risk.

The growing availability of open-source tools and managed platforms has certainly lowered the barrier to entry, but it has simultaneously raised expectations on throughput, observability, and user experience.

This article outlines an end-to-end serving architecture designed to meet low latency constraints while maintaining high availability and scalability. My focus here is practical: what works, what scales, and what

needs constant observation. Just like building a robust data governance framework is a journey, so is architecting for real-time AI inference. It's about building a foundation that will power your digital strategy.

Let's start with the orchestration layer. The first crucial decision you face when venturing into real-time AI inference is whether to manage the serving infrastructure yourself or to leverage a managed solution. This choice isn't trivial; it depends heavily on your organization's operational maturity, existing compliance requirements, and, critically, your performance constraints.

For many enterprise scenarios, managed model serving from platforms like Databricks Mosaic Model Serving, Azure ML Studios Endpoints, or Amazon SageMaker Endpoints provides a viable balance. These services abstract away a significant portion of the underlying infrastructure complexity. They handle automated scaling, SSL certificates, A/B testing, and version control, freeing your teams from much of the operational burden. They're typically suitable for workloads that can tolerate latencies in the range of ~50-150ms, offering a streamlined path to production without demanding deep DevOps expertise. These platforms often allow for automatic scaling based on demand, which is particularly beneficial for fluctuating AI workloads, just as cloud-native infrastructure is for a retail company during a holiday sale.

However, for organizations that demand stricter control over latency and require more aggressive optimization, teams often turn to self-managed solutions built on Kubernetes with KServe. KServe (formerly KFServing) is a Kubernetes-native serving framework that simplifies the deployment of machine learning models. It's a powerful tool, offering advanced features like scaling to zero (saving costs during idle periods), auto-scaling based on various metrics (requests per second, concurrency, CPU/GPU utilization), version management, and sophisticated traffic management (e.g., A/B testing, canary rollouts). KServe seamlessly integrates with various inference servers like TFServing, TorchServe, and Triton Inference Server, providing immense flexibility for different model types. The trade-off here is increased operational overhead; managing Kubernetes and its ecosystem requires a dedicated and skilled team. But for mission-critical applications where every millisecond counts, the superior customization and control it offers can be invaluable.

I would like to emphasize the steep learning curve or the need for strong SRE/DevOps maturity when adopting KServe. However if you are into the millisecond tier, you likely know that already.

In very specific edge use cases, teams might even bypass traditional orchestration altogether, but I will not get into that at the moment

Once you've chosen your serving framework, the next critical step is architecting the computational backbone for your real-time inference. This dictates both performance and cost. Your serving infrastructure must align directly with your model size, expected Queries Per Second (QPS), and, of course, your target inference latency requirements.

The debate of how much of the infrastructure will be "on prem" is a lengthy one with its own intricacies. I will not delve into that at the moment, but I want to highlight its importance: We are talking about GPU vs CPU, Horizontal vs Vertical Scaling, Predictive scaling, cooldown periods, load balancing, concurrency management and so on.

I want to emphasize that building out the right infrastructure for AI is not a one-time project; it's an ongoing journey of evaluation, adaptation, and optimization. Your teams must constantly discuss it, preferably in a "chapter" or "guild" model. As CTOs and CDOs, our role is to ensure this foundation is not only robust and efficient today but also agile enough to embrace the innovations of tomorrow.

Then, it is APIs time!!! The API design defines how your applications interact with the deployed AI models. It's the handshake between your front-end experience and your intelligent backend. This interface must be stable, fast, and, ideally, backwards-compatible to ensure seamless updates and integrations.

RESTful APIs are the workhorse of the modern web and are widely adopted due to their simplicity, statelessness, and broad tooling support. They're perfectly suitable for many real-time inference scenarios, especially when request payloads are relatively small and individual predictions are independent. Most data scientists and developers are familiar with exposing models as REST endpoints, and libraries like Flask, FastAPI, or Node.js frameworks make this straightforward.

However, for ultra-low latency, high-throughput, and streaming scenarios, gRPC presents a compelling alternative. Built on HTTP/2 and Protocol Buffers, gRPC offers several significant advantages:

Binary serialization: Protocol Buffers are significantly more efficient than JSON for serialization, leading to smaller payload sizes and reduced parsing overhead. This might seem like a minor detail, but at scale, these savings translate to substantial performance gains.

Multiplexing: HTTP/2 allows multiple concurrent requests over a single TCP connection, drastically reducing connection overhead. In a high-volume scenario, this means fewer handshakes and more efficient use of network resources.

Bidirectional streaming: This is critical for applications requiring continuous data exchange, such as real-time voice transcription, conversational AI, or the continuous updates common in gaming. Imagine a continuous stream of game telemetry being fed to an AI model for real-time anomaly detection or dynamic difficulty adjustment.

For finance, and gaming applications, where continuous updates or complex interactions are prevalent, gRPC can undeniably provide a performance edge over REST, often reducing end-to-end inference time a considerable amount.

Beyond the protocol, consider your batching vs. streaming strategy. In many real-time contexts, single-record inference is standard—you need a prediction for one action, right now. However, in other recommendation or personalization services, batching inference requests can significantly improve GPU utilization at the cost of some latency trade-offs. Beware of jitter or queuing delays and remember It's a balance between individual request latency and overall system throughput.

Finally, prioritize payload size and serialization. Keep your payloads lean; avoid sending unnecessary metadata. Use efficient serialization formats like Protocol Buffers or Apache Avro to minimize transfer overhead. Every byte counts when you're pushing for sub-100ms. And just as importantly, ensure your API is discoverable and well-documented.

You likely know me at this point: Data Governance and Documentation is an obsession of mine, so keep in mind that a well-maintained data dictionary isn't a bureaucratic exercise; it's the Rosetta Stone for your organization, especially when it comes to understanding how to interact with your AI services. Not to mention future-proofing development, avoiding refactoring and debugging.

My last point is about the silent killer of real-time performance: Network latency. Even with the fastest models and most optimized servers, if your data has to travel across continents, minimizing round-trip time (RTT) is absolutely critical, especially when targeting such stringent end-to-end latency.

While primarily known for static content, CDNs (Content Delivery Networks) can be cleverly leveraged to cache model artifacts or frequently requested inference results closer to the user. This reduces download times for model weights and overall latency for repetitive predictions. Think about semantic caching at the edge, where frequently requested model outputs are stored closer to the user, similar to how content is delivered faster from a local server. This approach isn't about replacing core inference but optimizing delivery of model-related assets or highly repetitive inference outcomes.

For applications with extremely stringent latency requirements (e.g., autonomous systems, real-time industrial control, or even specific gaming scenarios like cloud gaming), Edge Computing is the answer. Deploying inference capabilities at the network edge, physically closer to the data source or end-user, drastically

minimizes round-trip times to central cloud data centers. This involves pushing model execution onto edge devices (like IoT gateways, smart cameras, or even local micro-data centers). This strategy not only slashes network latency but also enhances privacy and security by processing sensitive data locally, reducing the need to transmit raw data over the network.

TLS Termination is another often-overlooked optimization. Terminating SSL/TLS connections at the load balancer or API Gateway reduces server-side processing overhead and allows the backend inference services to focus solely on prediction. Furthermore, using modern protocols like HTTP/2 or QUIC at the load balancer level can significantly reduce handshake overhead and improve multiplexing, further shaving off precious milliseconds. There is a pitfall I must warn you though in regulated sectors (e.g., finance, healthcare), terminating TLS at the load balancer must comply with zero-trust and end-to-end encryption policies. This may restrict TLS termination options.

Finally, consider the impact of your model itself on network efficiency. Model Distillation & Quantization are powerful techniques to create lightweight models. Distilled or quantized models (e.g., INT8) offer several benefits: faster serialization, lower memory footprints, and smaller binary sizes. These smaller models are inherently more network-efficient, requiring less bandwidth to transmit and load, contributing to a snappier overall experience. This isn't just about technical optimization; it's about delivering a seamless user experience, which ultimately fuels business value.

Advice for Teams Implementing AI

Real-time inference forces a powerful convergence of disciplines: ML engineering, backend infrastructure, and DevOps. In my experience, success depends far less on which specific model you choose and much more on how you serve it.

If you're embarking on AI implementation in your organization, start by gaining a crystal-clear understanding of your specific latency budget and reliability targets. Don't guess; measure. Then, choose serving frameworks and infrastructure that are designed to make performance observable and reproducible. Design your serving APIs with future integration in mind, prioritizing lean payloads and efficient serialization.

And always, always plan for failure. Whether that's an unexpected spike in demand, a model crash, or upstream feature drift—anticipate these scenarios and build resilient mechanisms like circuit breakers or rules-based fallbacks. Remember my discussion on the Human Factor: It's not about hunting unicorns, but about building balanced teams that can tackle these multifaceted challenges.

Don't treat model deployment as the finish line. It's truly the beginning of a living, breathing system that must adapt, scale, and stay accountable. The possibilities that AI unlocks are immense, but realizing that potential hinges on the strength and intelligence of the infrastructure that supports it. By prioritizing modularity, flexibility, efficiency, and strategic alignment—and by leveraging modern data technologies designed for these challenges—we can empower our organizations to deploy AI agents that truly transform our businesses, adapting to evolving needs without overwhelming our budgets or our teams.

What are the biggest real-time inference challenges you're facing in your AI journey? How are you approaching this balance between innovation, cost, and future-readiness? Let's keep the conversation going.

ARTICLE PUBLISHED IN 17 JUN 2025

Last week I had the opportunity to gauge the main events happening in Europe around AI and Technology: London Tech Week and VivaTech in Paris.

This year the atmosphere in both events was electric, but it wasn't just about the usual tech optimism. The message was clear and seismic: we're no longer talking about AI as just another tool. We're in the first days of a new industrial revolution, and AI is the foundational infrastructure upon which everything will be built.

For my fellow leaders, the takeaway is urgent and direct: The possibilities for reinventing your business are limitless, but this transformation demands leadership from the top. CEOs and CTOs, this is our moment to lead the charge.

Perhaps the most powerful concept from the events was this: the new programming language is human.

You no longer need to be a deep coder to "program" an AI. You just need to be able to ask clear, thoughtful questions. This democratizes technology on an unprecedented scale, unlocking latent talent across your entire organization.

This puts a lot of pressure into tech teams who should provide the infrastructure supporting the AI Agentic Ecosystems, fostering data governance, data visibility and championing data literacy to get data democratization and the transformation of the culture towards Data and AI

How are you and your leadership team preparing for this shift? What's the first step you're taking to build your own "AI Factory"?

#AI #DigitalTransformation #Leadership #CEO #CTO #Innovation #FutureOfWork #DataScience #VivaTech #LondonTechWeek

ARTICLE PUBLISHED IN 10 JUN 2025

The Human Factor in Data and IA

Welcome back. Over the last couple of weeks, we've covered a lot of ground together. We started by tackling the core challenges of getting your data house in order, moved on to the foundational infrastructure needed to support real ambition, and most recently, we delved into the bedrock of data governance. We've talked tech, we've talked pipelines, we've talked strategy.

But let's be direct: none of it matters without the right people.

You can have the most elegant Lakehouse architecture, the most pristine data models, and a governance framework that would make a regulator weep with joy. But if you haven't addressed the human element—the teams who use the tools, the talent you're trying to attract, and the cultural transformation required to make it all stick—you've built a high-tech engine that's going nowhere.

This brings us to what I consider the most critical element in any data transformation program: **The Human Factor**. This is where the magic happens, but it's also where the majority of data and AI initiatives ultimately stumble.

Data science, at its core, is a world of advanced mathematics, statistics, and uncompromising logical thinking. These are not, traditionally, the strongest traits you find in the people who excel at building client relationships, crafting compelling marketing narratives, or leading sales teams. Yet, in today's business landscape, we're telling everyone they need to change their DNA and become "data-driven."

This creates a fascinating, and frankly, dangerous paradox. On one hand, there's a genuine, widespread enthusiasm to understand and deploy AI. Everyone wants to be part of the future. The downside? This can breed an "ends justify the means" mentality. In the rush to get quick, visible results, teams often cut corners, bypass governance, and build solutions on shaky ground. It's a sprint for immediate validation in a world where, ironically, nobody believes anything will last long anyway.

But as leaders, our job isn't to chase fleeting wins. It's to build a lasting capability. And that begins and ends with people. Based on my experience in the trenches, the entire challenge boils down to three core areas: Teams, Talent, and Transformation.

The Teams Conundrum: Breaking Down the Tower of Babel

One of the most persistent and damaging challenges in any data project is the failure of cross-functional teams to truly collaborate. It's less of a gap and more of a chasm. We bring together data scientists, data engineers, business stakeholders, IT security, and marketing experts, and we expect them to seamlessly create value. More often than not, it looks like a modern-day Tower of Babel run by Jira Tickets.

Each group speaks its own jargon, operates on its own incentives, and carries its own biases.

While these groups don't share a common language or a unified understanding of the project's goals, the results are predictably messy: data silos deepen, misunderstandings fester, and projects either crawl to a halt or deliver a final product that, while technically sound, completely misses the business mark.

I once worked with a major retail company that was incredibly excited to launch a new AI-powered recommendation engine. The data science team spent a few sprints building a technically magnificent model. The accuracy was off the charts. When it was deployed, however, engagement tanked.

Why? The conversations with the people on the shop floor were protocolar, basically the data team interviewing for requirements rather than engaging in the nuance of the customer journey and the sales journey (two very different beasts, by the way). Don't get me wrong the model was a work of art, but it was solving a different problem. When we got the technical people to actually spend a couple of days on the field, accompanying the sales team, having lunch together, talking to clients, using the legacy order app which the sales field officers had to cope with, everything changed!

That's how I believe we fix this! It's not about forcing everyone to become a data scientist. It's about building bridges and this is what have worked for me:

I always start by creating dedicated "*translation layers*". In my teams, we've had immense success by formally establishing roles like Data Product Owners who operate as Analytics Translators. These people are bilingual—they can speak the language of business and the language of data. They sit in on sales meetings and then turn around and translate that context into clear requirements for the engineering team. They are the essential connective tissue.

Nobody understands the whole complexity of a major enterprise project, that's why *truly cross-functional teams* are paramount. I don't mean a weekly check-in meeting. I mean putting a data scientist, a product manager, a marketing expert, and an engineer into the same virtual (or physical) "pod," with a shared objective and shared accountability. When they are all responsible for the same outcome—not just their siloed part of the process—the conversations change dramatically. Suddenly, the engineer starts to care about the marketing campaign's click-through rate, and the marketer gets interested in the quality of the data feeding the model.

I risk being repetitive about the importance of communication but please double down on the *shared language and vocabulary*. Not in the literal sense of everybody speaking latin or something else. It is about my favorite topic: governance. A well-maintained data dictionary isn't a bureaucratic exercise; it's the Rosetta Stone for your organization. When "Active Customer" is defined, documented, and agreed upon by everyone, it

eliminates a whole class of miscommunication. It builds the foundation for trust and a common understanding that allows for effective collaboration.

The Talent Question: It's Not About Hunting Unicorns

This second great human challenge is the talent gap in data & tech. The data and AI landscape is evolving so fast that organizations are in a perpetual state of catch-up. We face a dual problem: a shortage of technical expertise and a critical lack of the soft skills needed to drive change.

The common reaction is to go "*unicorn hunting*"—trying to find that mythical individual who is a world-class statistician, an expert coder, a brilliant communicator, and a savvy business strategist all rolled into one. Let me be clear: these people barely exist. And if you find one, you probably can't afford them.

The real challenge is twofold: cultivating the talent you have and overcoming the deep-seated resistance to change. For many employees, the push for AI and automation isn't exciting; it's terrifying. It comes with a fear of redundancy, a discomfort with leaving behind familiar tools and processes, and a genuine lack of confidence in their ability to adapt. As leaders, if we dismiss this as simple stubbornness, we've already lost.

I saw this firsthand at a large insurance firm that wanted to use AI to help with claims processing. The most experienced claims adjusters—people with many years of invaluable institutional knowledge—were the most resistant. They weren't anti-technology. They were afraid that a black-box algorithm would miss the subtle human nuances they knew were critical to spotting fraud or identifying a vulnerable customer. They feared their hard-won expertise was about to be rendered obsolete.

The project only succeeded when we changed the entire premise. Instead of trying to reinvent them, we reframed the initiative to empower them. We turned them into "AI Supervisors." Their job became to train, validate, correct the AI models, and provide fundamental feedback on the new systems, getting financial prizes for successfully deployed model versions. We leveraged their deep expertise to make the machine smarter. This did two things: it dramatically improved the model's real-world accuracy, and it turned our biggest skeptics into our most passionate champions. They became part of the development pipeline, not targets for replacement..

Addressing the talent challenge requires a more thoughtful approach than just posting a job description. This is what worked for me:

First, **stop hunting unicorns and start building balanced teams**. You don't need every person to be good at everything. You need a team where skills are complementary: a great coder paired with a great communicator; a deep-thinking researcher paired with a pragmatic project manager. The team, as a whole, becomes the unicorn.

Second, **invest in transformation, not just training**. One-off workshops on Python or VectorAI are not enough. You need to build a genuine culture of continuous learning. This means giving people the time and resources to learn. It means creating dual career paths, so your most brilliant senior data scientist can continue to grow as an individual contributor without being forced into a management role they don't want. And critically, it means leaders must publicly celebrate learning and experimentation—especially when an experiment proves a long-held belief was wrong.

Finally, **recognize and reward the value of data roles**. If you treat your data team like a back-office cost center, don't be surprised when your best talent leaves for a company that treats them like a strategic asset. Fostering a culture where data expertise is visibly valued, respected, and integrated into strategic decision-making is the most powerful retention tool you have.

The Real Transformation: It's About Changing the Corporate DNA

This brings us to the ultimate human challenge: cultural transformation. I've said it time and time again, and I'll keep saying it until it sticks: **TECHNOLOGY FOLLOWS CULTURE**. You can spend millions on the most advanced data platform on the planet, but if your company's DNA is still wired for gut-feel decisions, you've accomplished nothing. The platform will become a ghost town, an expensive piece of shelf-ware.

Changing the DNA of a company is not something a Third-Party contractor, nor a Senior Manager nor a Director nor even a CTO can do. This is something that has to be driven by the CEO of the whole organization, because EVERYONE will have to change their mentality, processes and ways of working. Getting the most competent technical people and never having time for discussing how to become data oriented is a recipe for failure.

Successfully embedding data-driven practices is not a technical problem; it's a change management marathon. It requires rewriting the unwritten rules of how your organization operates. It means challenging legacy decision-making habits, building trust in automated systems, and getting genuine buy-in from every single level of the company.

The most powerful tool for cultural change is communication, specifically the celebration of small, concrete wins. Remember my point from the governance article about shouting your successes from the rooftops? This is why it matters. When you roll out a new self-service dashboard for the finance team and they use it to identify a significant cost-saving opportunity in the first week, you need to make that story legendary. Share it in the company all-hands, write about it in the newsletter, have the CFO talk about it.

This does more than just make the finance or commercial teams feel good. It shows everyone else, in tangible terms, that this new "data-driven" way of working isn't just more work—it delivers real, recognizable value. It proves that change isn't something to be feared, but something to be embraced.

This also requires deep empathy from us as leaders. We have to meet people where they are. We can't expect the entire sales organization to suddenly become experts in statistical modeling. But we can give them a simple, intuitive AI-powered tool that scores their leads and helps them focus their time more effectively. We empower them within their existing workflow, rather than demanding they adopt a completely new one.

Technology, in many ways, is the easy part. The real challenge—and the greatest opportunity—lies in empowering our people to harness it.

I would love to learn what are the biggest human-factor challenges you're facing in your organization? How are you bridging the gap between your technical experts and your business leaders? Let's continue this vital conversation.

ARTICLE PUBLISHED IN 03 JUN 2025

Data Governance: The Bedrock of Trust and Value

Welcome back to our ongoing conversation! Over the past month, we've delved into the critical foundations of successful data strategies I've experienced. We started by tackling the complexities of Infrastructure, the essential bedrock for any data-driven ambition. Then, we explored the nuances of building and maintaining that robust technological backbone. Now, as the third part of this series, the time has come to address an equally vital, yet often misunderstood, component: Data Governance.

For many, it conjures up images of suffocating bureaucracy, endless meetings, and a surefire way to kill innovation. It sounds like a massive, painful thing you just have to endure. To me this could not be further from the truth

Good governance cultivates trust at every single level of your organization allowing the “data driven” mentality to flourish. Your data analysts produce insights because they know the underlying data is sound. Your product teams confidently develop innovative features because multiple data sources indicate success. Your customers, critically, expect you to handle their personal information with the respect and responsibility it deserves. And your leadership team trusts that the business is operating in full compliance with ever-evolving regulations like GDPR, HIPAA, LGPD, PIPA or EU AI law.

But here’s the thing, and I’ve been repeating time and time again across different industries and cultures: Data Governance is not some static checklist you power through. It’s a living, breathing you cultivate. It’s not about locking your data in a cage; it’s about building a robust framework of trust around it. Let me tell you, trust isn’t just important – it’s pure gold. Good governance is the mint that produces it.

One of the toughest pills for many leaders to swallow is that data governance is fundamentally a journey, not a destination, it is a never ending process like quality obsession for a manufacturing operation. Business leaders, quite rightly, are wired to think in terms of clear goals, hard deadlines, and projects that get marked “complete.” They want to know when the “governance initiative” will be ticked off the list so they can move on. This is where the core misunderstanding lies. Data governance, by its very nature, will always be incomplete because your business is in a constant state of flux. New data sources emerge, regulations shift, products get launched, customer behaviors evolve. To imagine you can create a static, one-and-done governance plan is to ignore this dynamic reality.

How do you show progress on something that doesn’t have a neat finish line? The crucial lesson I’ve learned, often through navigating choppy waters, is that you absolutely *must* break the journey down into tangible, measurable milestones.

And your most powerful ally in this? Communication. Crystal clear, consistent, and celebratory communication. You have to shout about every small win. Did the team successfully define the data domains for the sales department? Announce it, explain its impact. Did you roll out a new, simplified data dictionary that clarifies the top 50 most critical KPIs for marketing? Share that success widely. These intermediate goals become the concrete achievements that demonstrate momentum, build crucial buy-in, and keep the energy alive for the long haul. They prove to stakeholders that while the journey is ongoing, the path is being paved with real, value-adding accomplishments.

Everytime I did that, the process advanced. Everytime I caved because that was not in the “culture of the company” to engage in this folly, data governance dragged on and its value perception decreased.

Good data governance framework provides clear answers to those fundamental questions: Who truly owns this piece of data? Who is ultimately responsible for maintaining its quality? How can it be used, and by whom? What are the ethical implications? Can I use this data? Can I trust it?

This last question is the most important: governance exists to build TRUST in the data so that *others* – people outside the core data team – can confidently use it. It also provides the rules of the road so everybody else can keep trusting the data while securely finding the new trustworthy applications and analysis created.

I strongly believe that the ultimate aim of any central data function is to get out of the way and become as invisible as electricity is in our daily lives. Thus the central data function has to empower the entire organization and foster the data and the data products that work, while focusing its expertise on the problems never faced before.

This is the very heart of data democratization and it is impossible to achieve without a robust data governance ethos.

When individuals across every department—be it marketing, sales, finance, or operations—can confidently access, interpret, and use data, algorithms, AI agents to make their own informed decisions, develop their own tailored reports, even run their own well-designed experiments, and incorporate data products into their applications, you've unlocked a truly powerful engine for widespread innovation and agility.

This generates growth, value and a deluge of new data that in turn needs to be governed making the data governance process fundamental.

Over the years, I've found that putting this philosophy into practice hinges on a few core principles that have proven indispensable time and again.

First, and I can't stress this enough, **communication and deep involvement are everything**. A data governance program cooked up in an IT silo, without genuine input from the rest of the business, is almost certainly doomed to fail. Everyone who generates data, or uses data to do their job – which, in any modern company, is pretty much everyone – must have a genuine buy-in. They need to understand the "why" behind it, not just the "what." When I was tasked with establishing a Data Governance Office in a previous role, our very first step wasn't to draft policies or select tools. It was to build a powerful coalition. We worked hard to get the CTO, the CFO, and even the Director of Compliance to co-chair the initiative. This wasn't just symbolic; it sent an unmistakable message from the top down: data is a shared strategic asset, and its governance is a collective responsibility, vital to our success.

Second, **embrace flexibility, especially when defining data domains**. This is often one of the most painstaking and time-consuming parts of the entire process. The natural temptation for a technically-minded team is to spend months, sometimes even years, trying to architect the "perfect," theoretically pristine set of data domains. What I've learned is that it's far more effective to develop a flexible, iterative approach. Come up with a solid, well-reasoned initial set of domains that can be carved out, expanded, or merged later as the business evolves and understanding deepens. The key is to communicate this initial framework to the wider company *sooner rather than later*. Get a working model out there. Let teams start using it. The feedback you'll gather from real-world application and the insights gained from how people actually interact with the data are infinitely more valuable than endlessly striving for an abstract, theoretical perfection in isolation.

Third, **don't reinvent the wheel; leverage established frameworks like DAMA (Data Management Body of Knowledge)**. Now, the DAMA framework is fantastic, but often not for the reasons most people initially assume. Its most powerful feature, in my experience, isn't necessarily its detailed technical prescriptions or its comprehensive lists of functions. While technically deep, its real power often lies in how it helps clarify roles and responsibilities for *non-technical* teams across these areas. This transforms governance from an "IT issue" into a genuine business partnership, making it tangible and actionable for those closest to the data's creation and use.

This approach provides a common vocabulary and a well-defined structure, assigning data ownership and stewardship roles and duties to the people who are closest to the data's creation, its context, its accuracy and its ultimate use.

The data technology team however has an immense responsibility to prioritize the approach order and translate to the business people how they fit into the eleven key knowledge areas encompassing in the DAMA framework. It is no small feat to break down concepts like:

- **Data Governance Policy:** Establishing authority, control, and decision-making over data.
- **Data Architecture:** The blueprint for managing data assets.
- **Data Modeling and Design:** Structuring and organizing data.
- **Data Storage and Operations:** Managing the physical and cloud-based storage of data.

- **Data Security:** Ensuring data privacy, confidentiality, and protection.
- **Data Integration and Interoperability:** Combining data from different sources.
- **Documents and Content Management:** Managing unstructured and semi-structured data.
- **Reference and Master Data:** Managing critical shared data like customer or product information.
- **Data Warehousing and Business Intelligence:** Providing data for reporting and analytics.
- **Metadata Management:** Managing data about data (e.g., definitions, lineage).
- **Data Quality:** Ensuring data is fit for its intended purpose.

In my experience, the best approach is to divide those into two categories: one business and process oriented and the other purely technical and data architecture oriented. Both running in parallel and reporting to a leader whose sole responsibility is the program implementation. Adopting a RACI approach makes things easier at the start and contributes to further break-downs.

The results were consistently transformative: increased departmental autonomy, more agile decision-making, improved regulatory adherence, and more reliable insights. This comprehensive approach didn't just fix immediate issues; it laid the foundation for a truly data-driven organization, trusted by all stakeholders and poised for innovation. Transforming your organization's relationship with data is a journey, demanding patience, constant communication, and a profound belief in the power of trust. By treating governance as a continuous improvement process and empowering everyone to participate, we build resilient, intelligent enterprises where high-quality, democratized data fuels sustainable growth.

The path isn't always easy, but a data-empowered organization is a destination worth striving for. What are your biggest governance challenges? How are you balancing control and enablement? Let's continue the discussion.

ARTICLE PUBLISHED IN 20 MAY 2025

CTO & CDO Infrastructure Challenges

In a previous article, I talked about the "three legs of the stool" for data and AI success: Data, Infrastructure Foundation, and Implementation. I got a huge response from that piece mostly around the foundation infrastructure supporting AI capabilities and legacy software development.

Encouraged by your feedback, today I'd like to double-click on that second leg, coming from my data-centric point-of-view of developing AI and Data products that not only generate insights and analytics but also are integrated into main software products, guiding customer journeys, offering promotions, and hyper personalization.... While keeping costs and latency down.

This isn't just about bolting on new hardware or spinning up a few more cloud instances; it's about a strategic architectural approach that has to be tailored to take our AI ambitions into account.

Based on my experiences leading numerous data and AI projects, successful technology stacks share key characteristics: modular infrastructure, resilient processing and pipelines, and a focus on future-proofing. Topics such as cybersecurity, metadata monitoring, data lifecycle monitoring, and application performance, while important, are intentionally excluded here for brevity as they build upon this foundational stack.

Building out the right infrastructure for AI is not a one-time project; it's an ongoing journey of evaluation, adaptation, and optimization. As CTOs and CDOs, our role is to ensure this foundation is not only robust and efficient today but also agile enough to embrace the innovations of tomorrow. It requires a relentless focus on modular design, careful management of cost levers, and, above all, a tight coupling between our data strategy and the overarching business objectives.

With that in mind, let's go through what I've seen gracefully handle the often-unpredictable and resource-intensive nature of AI workloads in a cost-effective manner.

Modular Infrastructure Architectures

When we talk about AI, especially training complex models or running inference at scale, the demand for compute and data processing can be immense and spiky. One minute you might be ticking along, the next, a new model training cycle kicks in, or a successful AI-powered feature sees a surge in user adoption, and your resource needs can go through the roof. This is where scalability isn't just a nice-to-have; it's the absolute bedrock.

Platforms like AWS, Azure, and Google Cloud offer incredible elasticity. They allow us to scale resources up or down almost instantaneously in response to fluctuating AI workloads. They also carry some level of risk as you might end up in an oversize contract that is expensive for the moment you are in your data journey, especially if your company talks a bigger game about data and AI compared to the actual level of automation and data products adoption.

The obvious benefits are avoiding massive upfront capital expenditures on hardware that might sit idle much of the time and the pay-as-you-go scheme that closely follows the actual demand. It is particularly beneficial to businesses heavily dependent on seasonal demand. Think about a retail company using AI for personalized recommendations. During a holiday sale, traffic might spike tenfold. A cloud-native infrastructure can automatically scale the AI inference services to handle this peak load and then scale back down when things quieten, optimizing both performance and cost.

Then there's the processing power itself. Large-scale data pipelines are the lifeblood of AI. We need to ingest, transform, and feed vast amounts of data to our models. This is where distributed processing frameworks like Apache Spark or Ray come into their own. These frameworks are designed to parallelize tasks across clusters of machines, dramatically speeding up data processing and model training. I've seen reports suggesting that effective use of these frameworks can cut infrastructure costs associated with these heavy workloads by as much as 30%. That's not pocket change.

Cloud platforms are not the only game in town. In my experience, most large organizations aren't starting with a blank slate. I have been fortunate enough to have worked for companies which have their own on-prem data infrastructure, that means taking into account legacy systems, existing data warehouses, and applications that are critical to the business. This poses the question of developing hybrid strategies and consistent integration work.

A hybrid cloud strategy often makes a lot of sense. It allows for a gradual migration of workloads and data, minimizing disruption to those crucial legacy systems. We can use the public cloud for its agility and scalability to rapidly prototype and deploy new AI agents, while sensitive data or deeply integrated systems might remain on-premise for a period. This phased approach balances innovation speed with operational stability but requires mastering integration on a different level. If you guys want to take the discussion through this route, reach me out and let's dive into the tricks and specificities of the hybrid approach

Future-Proofing an Infrastructure which evolves with AI

Any infrastructure that's cutting-edge today could be outdated tomorrow if it's not designed for flexibility. This is particularly true with the AI landscape evolving at breakneck speed and new algorithms, tools, and frameworks emerging constantly. Future-proofing isn't about predicting the future; it's about building an architecture that can adapt to it.

I'm a big proponent of Lakehouse architecture. As I've mentioned in previous discussions, most organizations are dealing with a mix of structured data (from databases, ERPs) and unstructured data (text, images, videos, sensor data). Traditional data warehouses excel at structured data, while data lakes are great for storing vast amounts of raw, unstructured data. A Lakehouse architecture, as the name suggests, combines the benefits of both, providing a unified platform that can support both batch processing and real-time AI agent training using all types of data. This versatility is key for many modern AI applications, from natural language processing to computer vision.

Containerization technologies like Docker and Kubernetes are also fundamental to architectural flexibility. Containers package up an application and its dependencies, ensuring it runs consistently across different environments – whether it's a developer's laptop, an on-premise server, or various cloud platforms. Kubernetes then orchestrates these containers, managing deployment, scaling, and operations. This approach not only ensures portability and avoids vendor lock-in (you're not tied to a single cloud provider's proprietary services) but also streamlines the deployment pipeline for AI models.

And as we build out these capabilities, an API-first design philosophy is paramount. By exposing data and functionalities through well-defined APIs, we simplify integration with new and emerging AI tools and frameworks. As the AI ecosystem continues to churn out innovations, an API-first approach ensures that our core infrastructure can easily connect to and leverage these new capabilities, maintaining agility and allowing us to adopt the best tools for the job without major re-architecting.

Finally I would like to bring up distributed technologies which are designed from the ground up to address many of these modern AI infrastructure challenges, like CockroachDB. It has been a long time since a vendor impressed me this much.

These distributed architectures are great for modern AI demands. You get seamless horizontal scaling across regions, automatic data rebalancing (no more manual sharding), and low-latency global access—all while preserving strong consistency. That's a major win for real-time AI systems. Plus, coupling this with elastic compute and tiered storage means you're not just scaling—you're optimizing resource usage dynamically, which is exactly what volatile AI workloads demand.

The Journey Continues

These foundational discussions around core requirements of scalability, resilience, data agility, and native integration with AI workflows, are just the beginning and we must consider cybersecurity and compliance concerns as we become any company's strategic enablers.

The possibilities that AI unlocks are immense, but realizing that potential hinges on the strength and intelligence of the infrastructure that supports it. By prioritizing modularity, flexibility, efficiency, and strategic alignment – and by leveraging modern data technologies designed for these challenges – we can empower our organizations to deploy AI agents that truly transform our businesses, adapting to evolving needs without overwhelming our budgets or our teams.

I'm keen to hear your experiences. What are the biggest infrastructure challenges you're facing in your AI journey? How are you approaching this balance between innovation, cost, and future-readiness? Let's keep the conversation going.

ARTICLE PUBLISHED IN 09 MAY 2025

CTO & CDO Challenges in Supporting Data and AI in Large Organizations

Many organizations today possess much more data than they can make sense of. The potential for artificial intelligence, data science, and machine learning to convert this data into value in operations is considerable. However, the journey from this potential to business outcomes with measurable effect is tricky.

Following the roadmap I discussed in my last article on things I learned in the trenches, in this piece I will touch on something professionals coming from a software development background face when they take over as CDOs or CTOs. Many technology leaders have developed software their whole careers without realizing how nuanced and specific data projects are. Now as AI and Data are blobbing their way onto the technology field, these leaders oftenly ask: "What should I focus on to support a true data driven push?"

As many of my colleagues, friends and readers know, my work experience in data transformations across various sectors has shown that success depends not only on technology. Strategy, people, and culture are

huge factors. I will expand on those later. Today I will discuss the three legs of the stool leaders will likely be standing on when implementing data products and AI for large organizations.

The first Leg: Data

Let me start with the data itself: Some organizations fall into the trap of trying to "Catalog the Universe" before even starting to engage the potential use cases and business value they could obtain from becoming data oriented.

Collecting, transforming, storing, and retrieving data are complex, costly and messy activities. Redundant types and often contradictory data, collected by different methodologies and under different definitions are the norm and a main source of cost, confusion and inefficiency. In large companies, data integration projects frequently face difficulties from data formats that differ and conflicting standards, especially when interacting with systems from an earlier time.

Therefore, establishing data quality, data consistency, and implementing governance with depth and breadth is a foundation step. Setting benchmarks for data quality, fully sanctioned and cared for by responsible data owners guarantee the organization has data that can be trusted by anyone. The technology area must establish and implement governance mechanisms that operate with automation, including data lineage tracking and data dictionaries with detail that is comprehensive, and assist in maintaining quality when operating at scale. These systems aid compliance with regulations such as GDPR or HIPAA.

This is such an important and underappreciated basic part of the job. I plan to write in detail about it in the future, specifically how these programs lead to data democratization, empowering business teams and facilitating the widespread adoption of AI across organizations.

The second leg: Infrastructure Foundation.

As the quest for taming the data deluge goes on, in my experience a continuous assessment and strengthening of the technological infrastructure that supports the whole thing must be carried on.

The scope of this work is wide, encompassing the various IT platforms that support both current business operations as well as the ensuing data applications: from data analysis, ML development and automation to AI agents integration, those applications inevitably add stress onto the overall infrastructure responsible for running apps, legacy systems, ERPs, and every aspect of a company's business.

I can tell you that you will be responsible for maintaining and improving what currently exists and works, while expanding it to perform crucial data driven activities like collecting old and new data, cleaning, organizing, development and testing. Once this is all over, then the infrastructure will consume, deploy and support all the delicious innovations and AI trinkets we love.

Hence, the infrastructure foundation shall support a multi structure and full-cycle data usage designed for AI automation, which will make the analysis and suggest the decisions (or even make the decisions autonomously). This is a game changer as the analytical environment has to be fully integrated with the operations environment, to measure, evaluate and improve the consequences of those decisions.

Most companies are not even there yet. Most still have in mind the analytical part of that data-driven arch: creating dashboards to support "better informed decisions". But even in those the game has moved from ingestion and dashboards to complex integrations, particularly with platforms from earlier eras, embracing new environments, flexible designs, modular components, and innovative architectures. In those companies technology infrastructure was designed to deal with structured data, but as digitalization grows, numerous data formats and unstructured data sources like video, conversations, mechanical sounds must be stored, processed and used in advanced algorithms and ai agents.

I favor architectures built from interchangeable components and solutions designed as a service for their interoperability and capacity for growth. This allows component reuse without waste, avoiding rework that incurs high expense. But how we utilize those architectures has changed tremendously in the last three years.

This discussion is far beyond the familiar on-premise versus cloud. It is about how to best use multiple environments to construct a data backbone that shows resilience as businesses expand and demand scaling infrastructure capabilities. Legacy IT systems might not satisfy demand and need to be replaced. Cloud-native architectures, serverless computing, and provisioning with automation offer flexibility but might become a vulnerability, a liability or both.

This topic as I expected has drawn a lot of attention and I got such tremendous feedback from my data architect friends and colleagues that I plan to get back to it in future articles. This also has raised the need of the technology area to embrace a data driven approach onto itself!

For now, lets move to the final leg

Third leg: Implementation Innovation, Scale, and Data Products

With data foundation and infrastructure progressing, attention moves to balancing innovation pursuit with steadiness in operations. It is expected that Chief Technology Officers guide digital transformation while keeping the lights on and the business operating seamlessly. This is a balance act that requires a keen eye on the future of the industry and the company, slicing technology adoption in stages, and non stop process optimizations. It involves embracing new tools, maintaining systems that function well, and retiring elements that are no longer current or may have become a liability in cost, security, or maintenance.

And yes, again, this requires a data driven approach, with the technology area generating data about itself, analysing and evaluating its own performance at set intervals, planning for capacity, finding bottlenecks and the costs imposed on the area and on the broader organization.

However this third leg is about something completely different from the technical topics covered so far. This is about dealing with the consequences of a successful journey until this point. This third leg is about developing people so they can reap the fruits of organized and comprehensive data and a powerful technology stack capable of making their most innovative business idea come true.

This is where I saw so many companies stumble.

At this point, companies are excited about the success and novelty of the technologies introduced, the power of data, the workshops, the internal stakeholder engagement, the encouraging results of MVPs and POC. That is when internal demand explodes as every single department in the organization wants to jump into the bandwagon and get the benefits of this data-driven digital transformation process.

It is time to scale from a central data function whose job is to carefully curate data and skillfully develop data products to an open and wide full scale transformation where data assets are tailored for, or adapted by individual business departments to new requirements, under standard frameworks of common governance. That's when data products are scaled up to volumes and specifications that far exceed the original ones sanctioned by the central data function.

While working on this leg CTOs and CDOs are required to provide training, keep a watchful eye for compliance, cybersecurity, support data and software engineers to learn non technical skills: from agile and customer centricity to empathy and text interpretation, while providing to salespeople, layers accountants, medical doctors, surgeons, marketing experts, cooks the technical training they need to develop their own data products, AI agents, in a responsible and effective manner.

In my experience those three legs must not be built apart from each other, but all at once, one supporting the other. Oftenly there will be different maturity levels for each of them and that is fine. Though they are NOT the

only aspects that guarantee the success of data projects, I hope that if you step up to a leadership role in data and technology, you will find these thoughts useful.

The stage is set for data democratization, data literacy, people engagement, data security, and strategic alliances to come into focus. I want to talk about them next time.

I would love to know what you think, what in your experience supports successful implementations, please let me know if I missed something or if I left something out. Let's keep this conversation going.

ARTICLE PUBLISHED IN 4 MAY 2025

May the Fourth be with you all! ✨

Like many of you, I absolutely love the *Star Wars* saga. Beyond the epic space battles and iconic characters, I often see powerful parallels between building that incredible universe and launching successful data projects right here in our own galaxy. Think about it – isn't the journey of deploying a new app or AI Agent as thrilling as the Death Star's trench run? I was reminded of that last night as I watched Andor's episode 6 (what a fantastic show) and I would like to share some thoughts about the franchise on this day:

This continuously expanding space opera's original trilogy started centered on Luke, Leia, Han, R2 and Vader. It told a compelling, core story before expanding outwards. Similarly, our most impactful data projects often begin by addressing a specific, critical need, delivering value quickly before scaling up. Paraphrasing Vader: *"If you only knew the power of the MVP"* we must always start with a focused vision that tackles a real, tangible problem.

And just as the Star Wars universe established its fundamental rules early on – how the Force works, the nature of hyperspace, the might of the Empire – strong data projects need solid governance from the outset. Defining these core principles, structures, and standards allows our data ecosystems to grow coherently and reliably. Paraphrasing Master Yoda: *"Data Governance is strong in this one"* . Indeed, strong governance does not hinder nor turn things "bureaucratic". It creates the foundation upon which we can build, innovate, and add new capabilities, ensuring everything works together harmoniously.

Neither Cassian Andor, Grand Admiral Thrawn nor Ashoka Tano were created by George Lucas, but they are fantastic characters whose stories make the whole thing richer, more enjoyable, more inspiring. As other authors and fans inserted their perspectives, co-creating plots and storylines and adding value to us all, *Data Democratization* comes into my mind. The same way the rebellion belongs to the whole galaxy *Great data projects do not belong to a single centralized team* . Those expand on their own, driven by user's needs, specific commercial goals and developed by data savvy padawans, knights and masters. It is our job as leaders to nourish these heroes and support them to scale effectively and confidently.

What parallels do you see between your favorite stories and the world of technology, data and AI? Let's share our insights and inspire each other to build data projects that are truly out of this world!

...And May the Force be with us... always!

ARTICLE PUBLISHED IN [APRIL 2025](#)

Okay, let's get straight to the point. We're constantly told that data is the new oil and AI is revolutionizing everything. It sounds great, like flipping a switch and suddenly having a smarter, faster, better business. But let's be real – it's rarely that simple. While data and AI are becoming as fundamental as electricity, translating that potential into real-world business value is where most organizations hit a wall.

My name is Erlon Rachi. For over two decades, I've been deeply involved in data transformations across various industries and three continents. I've seen the promise, but I've also seen the pitfalls firsthand. Remember that sobering statistic? Something like 87% of data science projects never reach production. Why? Because success isn't just about technology; it's about strategy, culture, and execution. It's about moving beyond simple reports to genuinely embedding data insights into the fabric of your business.

That's why I'm kicking off this series of articles. My goal is to share practical, on-the-ground insights – lessons learned from years in the trenches – to help you, your teams, and your companies navigate this complex landscape. This first piece is an invitation: let's explore together how to genuinely unlock the power lying dormant in your data, paving the way for meaningful automation, machine learning, and AI.

My plan is to engage in a conversation with you, sharing my experiences and learning about yours. I plan to organize these conversations by topics which are key elements of success, positively influencing and supporting each other, carrying the lessons that we've learned along the way. I am counting on your feedback to gauge how deep to dive into those topics, either exploring code, case studies, libraries, books, references and conversations with the friends I made throughout my journey.

The goal is to help anyone to achieve tangible business results through data and AI projects by discussing what supports them. Today, however, I want to lay out these topics and how I see the conversation evolving.

I will start with The Data and its supporting Infrastructure Challenge: "More Isn't Always Better" and the recognition that we're swimming in data and the pool deck is not for free, nor was it built by nature. Collect, transform, store, retrieve, identify possible value in that data, verify that value, these are not only computationally intense operations but also costly ones. Building this foundation of structure, ingestion, platforms and frameworks has its challenges and we could stay discussing the pool and the water for ages, but we must not miss the point: this pool and the water in it has a purpose and we must understand it. Are we hosting the Olympic swimming competition or a pool party?

I will move on towards the necessity to find beauty in the chaos and that's when Data Governance shines. It isn't red tape; it's the framework that ensures your data is accurate, consistent, secure, and ethically handled. Who owns the data? Who can use it? How is quality maintained? Good governance builds trust – trust in your insights, trust from your customers, and trust that you're complying with regulations. In the age of data and AI, I believe trust is gold!

I also want to talk about a conviction I hold strongly: technology follows culture. You can invest heavily in platforms and explore castles on the clouds, but if your organization doesn't embrace data literacy, foster collaboration, and empower data-informed decisions, you'll hit a ceiling.

It deserves a topic of its own, but it is a key element present in every other one: the Cultural change and the role of leadership, not only changing minds on the field, and in our stakeholder's departments, but also in the technology and data areas. This is about shifting from simply reporting what happened to using data to understand why, predict (or at least give a great hunch) what's next, and make the decision about what to do the most educated possible.

Leadership also plays a vital role in championing the creation of an environment where experimentation is encouraged (and yes, that means learning, analysing the data, from occasional failures). A true data-driven

culture is the engine of sustainable innovation. And nothing helps this more than embracing agile, experimentation and scale!

This is what Data Digital Transformation is all about, as it requires encouraging curiosity and critical thinking with data across all levels. It means breaking down silos so business experts, tech teams, and analysts work together, speaking a common language.

This also applies to identifying the core "Burning Business Question" – the specific problem you're trying to solve – data efforts often become costly exercises in exploration with no clear destination.

I plan to dive into the meta portion of the story: acknowledging how little data areas use KPIs themselves, how to use models internally, self improvements, and scaling up our very own impacts. I will also write about the pitfalls regarding training and onboarding, the relentless and rewarding stakeholder management process, and the link between strategic thinking and budget planning and execution. ,

Join the Journey

Transforming into a truly data-driven organization is a journey, not a single project. It requires a blend of strategic vision, technical know-how, cultural change, and unwavering focus on delivering measurable value.

My passion lies in helping organizations navigate this journey. Through this series, I'll share practical frameworks, real-world examples, and lessons learned on topics ranging from governance and team building to ethical AI and specific implementation strategies.

The future belongs to organizations that can intelligently harness their data. It's about building not just smarter systems, but smarter, more adaptive, and more trusted businesses.

I invite you to join me in these upcoming posts. Let's explore how to move beyond the buzzwords and turn your data potential into a powerful, sustainable strategic advantage. I look forward to sharing more and hearing your thoughts.

Last I would like to thank the support and inspiration I got to embark on this series: Mark Rodseth , Carolina Wosiack Naomi Garratt Andrew Beckett Nikos Anastopoulos Henadzi Dubin Breno Spadotto Miguel Frasilho Jamie Norrington Prof. Dr. Marcos Santos Thank you very much!

ARTICLE PUBLISHED IN MARCH 2025

Implementing AI solutions: The Practical Playbook for Innovators

Let's face it—AI is the buzzword of the decade, and everyone is rushing to integrate it into their business. But here's the rub: about 87% of data projects never make it to production, and only 27% are considered profitable. So, how do you avoid becoming part of the gloomy statistics?

Having navigated through hundreds of projects across multiple industries, I've seen what makes or breaks an AI initiative.

Whether you're a scrappy startup or a seasoned enterprise, here are the crucial steps you need to get right to ensure your AI solution doesn't just survive—it thrives.

You can't build a solution without a clear problem. Successful AI projects always start by asking: "What's the burning business question?" AI isn't a magic wand; it's a powerful tool that needs a clear direction. Setting the purpose of the AI agent you want to develop or deploy might be the most difficult one in the whole journey.

Next, truly embrace agile. The faster you iterate, the quicker you learn. Enterprises often get bogged down in infrastructure concerns—storing data, processing algorithms, governance, security, and compliance. Meanwhile, startups typically bypass this complexity by leveraging cloud-based solutions, focusing their energy on experimenting with compelling products or services. Breaking down the roadmap, trying out MVPs help you cut through the maze. The goal is to get something done, not build the best AI agent in a single project. Work iteratively!

None of this works however if the right people are not involved. Not only the right technical people who are developing the AI, or integrating and deploying it, but also the stakeholders, customers, peers.. AI projects hinge on team capability and a data-driven culture. Skills are vital, but curiosity and adaptability are even more critical.

This is, however, only the start of a process as creating the AI solution itself is just your first punch. Sure, it feels great to develop an algorithm that solves a complex problem, but your second punch—scaling your solution—is where true impact lies.

This second punch requires scaling up and addressing issues like reliability, traffic volumes, security, compliance and, most important of all, measuring the costs and the effectiveness of the AI agent addressing the Burning Question that has motivated its development.

A successful implementation then is one that has passed the three main filters: Cost, considering not only the initial development costs, but also factoring in AI tokens, infrastructure usage, maintenance, and scaling expenses. Time to Scale: Realistically assessing how quickly your MVP can evolve into an operational asset across the organization and finally Effectiveness, measuring how effectively your AI solves the core Burning Question..

Whether your company feels more like a startup or an enterprise, your success with AI boils down to clarity of purpose, agile processes, and people who understand, embrace, and innovate with data.

1. The AI Implementation Journey

From your experience, what are the most critical first steps for a company looking to implement AI solutions successfully?

The very first step towards implementing AI is leveraging human intelligence in the company: That means having data savvy people in the team who are able to identify the low hanging fruits, meaning which problems, bottlenecks or inefficiencies yield the most impact, considering that they can be addressed by available data. There are many flavors of AI solutions: from GenAI that creates codes and marketing campaigns to machine learning algorithms that recommend the next product to be offered. Choosing which to start implementing requires human intelligence over the company's goals.

Second is embracing experimentation: testing those solutions in small markets, or in limited time periods, learning from the results, which is just another way to say that companies need to gather data and analyse it against company objectives. Experimentation is the process of accruing the impact each solution has brought, understanding what works and what does not work.

Third is consistency in scaling up and iterating: that means that whatever solution that works must be expanded to cover the whole business while they need to continue to be measured for efficiency and improved

This is a never ending cycle, based on data, experimentation and innovation, aligned with business success. As companies go through it, more complex and sophisticated AI solutions can be deployed.

2. Identifying the Right Problem

AI is only as good as the problem it solves. How can organizations ensure they are applying AI to the right challenges?

Before starting companies must run an assessment of the existing challenges, capabilities, investment levels and benefits being pursued. The objective is to balance technical feasibility with business impact. The candidate problems must align with company core business strategies: focusing on cost-control and efficiency if the company operates in a resource intensive industry or focusing on scale when operating in large horizontal markets.

My favorite framework is an AI opportunity matrix that maps potential use cases against two axes: strategic value (revenue impact, customer experience) and implementation complexity (data readiness, technical complexity, financial ability, current stage of data development)

3. Scaling AI Beyond the Prototype Stage

Many AI projects never make it past the MVP phase. What are the biggest barriers to scaling AI solutions, and how can companies overcome them?

In my experience, there are three big barriers: First is trying to utilize the same AI algorithm or ML methodology developed for the MVP in the scaled up version. It is the equivalent of knowing how to prepare pasta at home and then working the same way to feed an army battalion. Even though the main course is the same, things run very differently in the kitchen. Different algorithms, codes, and tech stack are needed to scale things up

The second barrier is cost: they differ greatly from MVPs into scale. Investment costs have to be factored in, as well operational costs, training, security, compliance, redundancy. P&L analysis is still king

Third is the integration of the AI solution into the overall customer experience. One must never forget that AI is part of the customer experience, it is not all of it. Any AI powered solution must fit perfectly into the customer journey and the overall experience. If AI introduces an element of oddity and weirdness that undermines other aspects of the company's offer, it might bring problems far greater than the ones it initially was designed to solve

4. The Agile vs. Enterprise Approach

There are two dominant approaches to AI implementation—enterprise-level infrastructure vs. agile experimentation. How should companies determine which approach suits them best?

It is not one OR the other, but one AND the other. The important thing is understanding WHERE companies are in the development cycle I discussed before. Fast agile experimentation is essential to identify the existing problems, costs, ROI, interactions with established business models, while Enterprise-level approach usually tackles the topics related to scalability, reliability, security, basically things that allow the MVPs to profoundly transform the business and unlock major returns.

5. Measuring AI's Impact

What are the most effective ways to measure the success of an AI implementation in terms of cost, scalability, and effectiveness?

This is fundamentally linked to the company expectations prior to the AI deployment which heavily influence the calculation of the return of the investment. When it comes to cost, it is important to consider the factors like the total cost of ownership (TCO), which includes data acquisition, model development, cloud/compute expenses, and maintenance and the reduction of operational costs (OCR) which take into account productivity increases, error correction, or process inefficiencies.

Scalability usually involves the creation of KPIs that measure performance like Inference Latency which is the time for AI to generate results as load increases, cloud costs per million requests, etc.

Effectiveness metrics is always measured by an internal metric that has to make sense for the leadership reflecting a balance between technical results, like how well a given model identifies fraud, and business results, like how much money was saved by flagging fraudulent operations.

6. The Role of Data in AI Success

High-quality data is the foundation of AI. How can companies ensure they have the right data infrastructure to support their AI ambitions?

First it is necessary to have an overkill infrastructure capable of cataloging the universe! That's why differentiating the ambitions from what is needed at this point is the company's journey is so important.

In my experience, one must always start working using scalable architectures that can handle the amounts of structured and unstructured data the company needs at that particular moment. Cloud-based data lakes, modern data warehouses, and real-time processing pipelines are essential to ensuring that AI models can access and process information efficiently. Without these core capabilities, AI projects often become bottlenecked by slow retrieval times, fragmented datasets, and rigid legacy systems that cannot adapt to evolving business needs.

Moving beyond infrastructure, a strong governance to maintain accuracy, consistency, and security is the often overlooked stepping stone that guarantees that data can be trusted and put to good use. Security and compliance are also part of the governance framework and both play a critical role ensuring AI systems not only drive innovation but do so responsibly.

Ultimately, AI success is as much about culture as it is about technology. A strong data infrastructure must be accompanied by a shift in mindset, prioritizing cross-functional collaboration, data literacy, and ethical considerations. Companies that foster a culture where business leaders, data engineers, and AI researchers work together will be better positioned to turn their data into a true competitive advantage.

The most successful AI implementations are those that align with business goals, integrate seamlessly into workflows, and are built with long-term sustainability in mind.

7. AI and Decision-Making

AI is often seen as a tool for automation, but how can it be leveraged to enhance strategic decision-making within businesses?

Algorithms focused on automation, efficiency and quick response belong to the type of AI and Machine Learning mankind has already put to use for the last 60 years. What is new about the current stage of AI is that it already is far more than an automation tool. It has become a powerful engine for strategic decision-making, capable of uncovering patterns, predicting trends, and guiding leaders through complexity with data-driven insights.

The challenge is to organize the flow of data, creating and analyzing vast datasets so AI can simulate different scenarios. AI agents and specialised analytics help businesses anticipate market shifts, mitigate risks, and optimize resource allocation with greater precision. It also enhances human judgment by surfacing hidden correlations and challenging biases, ensuring that decisions are based on evidence rather than intuition alone. But for AI to be truly transformative, companies must embed it into their strategic processes, fostering a culture where data-driven insights shape long-term vision and competitive advantage. When leveraged effectively, AI becomes not just a tool for efficiency, but a trusted partner in shaping the future of business.

8. The Ethical Challenges of AI Implementation

As AI adoption grows, so do concerns about bias, transparency, and accountability. What steps should organizations take to ensure ethical AI deployment?

Bias in AI arises basically from two root causes. The first one is when models learn from skewed or incomplete data, leading to unfair outcomes that can reinforce the bias contained in the datasets. The second one emerges from humans who are interacting with AI carrying their own biases into the models by the prompts, flows, or selections they provide to the algorithms powering AI.

To combat this, companies must implement rigorous data governance practices, regularly auditing datasets for representativeness and fairness. Techniques such as bias detection algorithms, adversarial testing, and diverse data sampling can help mitigate unintended discrimination.

But technology alone is not enough. Fairness and ethical behaviour must be part of the company culture and training. Actions like establishing interdisciplinary review teams, bringing together ethicists, legal experts, and domain specialists to scrutinize AI systems before they are deployed at scale must be part of the toolbox.

Transparency and accountability are equally critical. AI models should not operate as "black boxes," where decisions are made without clear explanations. Instead, AI development must prioritize explainability by using interpretable AI techniques, such as counterfactual analysis, to clarify how models arrive at their conclusions.

In my opinion, this approach which is favored in Europe, the UK and is at the foundation of the open-sourced AI community is the most sensible one.

9. Common AI Implementation Pitfalls

What are the most common mistakes companies make when implementing AI, and how can they avoid them?

One of the most common mistakes companies make when implementing AI is treating it as a plug-and-play solution rather than a strategic investment that requires careful planning and integration. Many organizations rush to deploy AI without first ensuring they have high-quality, well-governed data, leading to unreliable models and poor performance. Others underestimate the need for cross-functional collaboration, leaving AI initiatives siloed within technical teams without input from business areas, domain experts, or end users.

Another critical misstep is failing to define clear success metrics and expecting immediate results without iterative refinement. AI is not a one-time deployment, it is the opposite of it! AI requires continuous monitoring, retraining, and adaptation as business conditions evolve and data used to model situations change, because society and business change.

To avoid these pitfalls, companies must first invest in AI literacy across all levels, ensuring leadership and employees understand how AI functions and its limitations. As this change mindset is going on, invest in a strong data foundation, ensuring their AI models are trained on clean, representative, and unbiased datasets.

Equally important is fostering collaboration between AI engineers, data scientists, business strategists, and frontline employees to ensure AI is designed with real-world needs in mind, establishing feedback loops, incorporating human oversight and performance audits to catch errors before they escalate.

10. The Future of AI Implementation

Looking ahead, how do you see AI implementation evolving, and what trends should businesses prepare for in the coming years?

AI implementation is poised to move from boardroom buzzword to an invisible, indispensable force running beneath the surface of everyday business operations. Companies will no longer ask, "Should we use AI?" but rather, "How do we solve this?"

I expect AI to shift from clunky, one-size-fits-all models to hyper-personalized, context-aware systems that adapt to individual users, industries, and even moods (yes, your AI assistant may soon know when you need coffee before you do). Businesses should brace for the rise of AI copilots not just automating tasks but actively enhancing human decision-making and helping us "get in the zone" in a seamless manner, making us more productive and reflective as we reason with AI to get better outcomes.

Ethical AI, explainability, and regulatory compliance will become non-negotiable, not just to avoid lawsuits but to build trust in an increasingly skeptical world. The future of AI isn't about replacing people; it's about augmenting them.

As the AI powered economy takes shape, I strongly believe that the currency of the future is TRUST. If your company can accurately interpret the vast amount of data being collected from the world, it will be trusted and therefore valued by customers, partners and the society.

ARTICLE PUBLISHED IN SEPT 2024

As an exec in data and tech, it's crucial to sync personal brand goals with company strategy, avoiding clashes. Balance technical expertise with the particularities of the company, respecting how data is created and its intended practical usage. Be clear and steady about the governance structure, supporting your team while

valuing honest feedback. Pushbacks are inevitable and when considered carefully either lead to a better governance rules or to the broader organization comprehending the need to adhere to the governance structure. Networking must be used wisely as it might lead to partnerships you must always keep in mind confidentiality and sensitivity of internal processes and intentions that generated the data environment you are in charge of governing.

ARTICLE PUBLISHED IN JULY 2024

Leaders must balance tailored communication with ethical considerations and authenticity. This is particularly important in the Tech Industry and becomes a must when interacting with people that do not have a technical background. It demands more than superficial adaptation but truly grasping where your audience's understanding and focus are, and the vision you want to carry. How can your messages resonate deeply without diluting your vision as a leader? For me these are the two key elements: First, ethical and engaging storytelling woven with relevant data that supports every aspect of the message: the goal, capabilities, risks and the immediate actions needed. Second, why should your audience care, what is in it for them, the reasons why they should engage and think about your message. This generates meaningful and effective feedback, fostering trust and impact.

ARTICLE PUBLISHED IN AUGUST 2024

We all build models of the world in our heads – it's how we make sense of things and predict what's coming next. But garbage in, garbage out, right?

That goes for our brains AND our AI! This awesome article by [Damien Benveniste, PhD](#) dives into the nitty-gritty of optimally sampled imbalanced training data. Because just like we need to read different viewpoints and connect with diverse people, AI needs a balanced input to avoid becoming a narrow-minded bot.

How do you deal with imbalanced data? If you don't have too much data and the imbalance is not too extreme, the typical way to deal with it is to simply reweigh the samples such that the loss function considers the positive and negative samples equally. When you have an overwhelming amount of negative samples, you may want to downsample them to minimize training latency.

But not all samples are equal! At TikTok, for example, for their recommendation engine, they use a non-uniform negative sampling scheme they developed with the University of Connecticut: <https://lnkd.in/gRsFSr2d>. They proved that optimal sampling of the negative class is done when giving more weight to samples with a higher probability of being positive (Theorem 3). This means that it is better to keep samples that are confusing for a model. This way, the model focuses on learning how to separate true positive samples from negative samples that look like positive ones.

Interestingly enough, this theorem also means sampling bias is a good thing! In ML applications, a model shows users some samples they are likely to engage with. When they don't engage with those, they become negative samples for the next training batch. That is sampling bias because only the samples with a high

probability of engagement ever get shown to users, and they never get the opportunity to interact with the "lesser" samples, so we never get signals for those.

By sampling the data, we bias the probability estimates coming out of the model, and they become meaningless. The model is not calibrated anymore. To fix that, they came up with a correction of the likelihood function to generate unbiased estimates of the model parameters and, therefore, the probabilities (see eq 5).

Practically, you follow this process to sample down negative samples:

- 1) Uniformly sample the negative class so that the data becomes balanced.
- 2) train a model with balanced data. They call it a "pilot" model.
- 3) predict the full data with that pilot model. You get an estimate of how much the model believes the sample is a positive one.
- 4) normalize that probability p by the average probability w and multiply by the sampling rate r : $r * p / w$
- 5) for each negative sample, pick a uniform random number u . If $u < r * p / w$, keep the sample; remove it otherwise. The greater p is, the more likely we will keep it
- 6) $r * p / w$ is the sampling probability. When training the model or predicting, correct the log odds using that probability.

ARTICLE PUBLISHED IN JULY 2024

A Data-Driven Executive's Perspective: Monitoring for Continuous Improvement in Data Governance Training

The article provides a comprehensive overview of the key considerations for monitoring a data governance training program. However, I believe it's essential to go beyond mere monitoring and emphasize the importance of **continuous improvement** based on data-driven insights. Here's my perspective:

1. Embrace a data-centric approach. The article mentions collecting data, but we need to move beyond basic metrics and delve into **advanced analytics**. Leverage data visualization tools to identify trends, patterns, and correlations between training elements and trainee performance. This allows for **data-driven decision-making** to optimize training content, delivery methods, and overall program effectiveness.

2. Foster a culture of feedback and iteration. The article suggests collecting feedback, but we should **actively encourage continuous dialogue** with trainees and stakeholders. Utilize surveys, focus groups, and open communication channels to gather **qualitative and quantitative feedback**. This ongoing feedback

loop enables **rapid iteration** and ensures the training program remains **relevant, engaging, and impactful**.

3. Prioritize transfer of learning and behavior change. Monitoring trainee performance beyond assessments is crucial. Track **real-world application** of learned skills and observe **changes in data governance behavior** within the organization. This ensures the training program translates to **tangible improvements** in data quality, compliance, and overall data governance maturity.

4. Adapt to the evolving data landscape. The article briefly mentions the need to update training for emerging technologies. We should **proactively anticipate** future trends and challenges in the data governance landscape. This involves **continuously scanning the horizon** for new regulations, technologies, and industry best practices, and **integrating these insights** into the training program to ensure it remains **future-proof**.

5. Cultivate a learning ecosystem, not just a training program. The article focuses on a single training program. We should view data governance training as part of a **holistic learning ecosystem**. Encourage **self-directed learning** by providing access to additional resources, knowledge-sharing platforms, and mentorship opportunities. This fosters a **culture of continuous learning** and empowers individuals to become data governance champions within the organization.

By adopting this data-driven and iterative approach, we can transform data governance training from a one-time event to a **continuous improvement engine**. This ensures our workforce remains equipped with the **knowledge, skills, and behaviors** necessary to navigate the ever-evolving data landscape and **unlock the full potential of data** as a strategic asset.

In my experience, data governance isn't just about firewalls and access controls. It's about building a foundation of trust around our data, transforming it from just locked-in numbers into strategic assets.

Sure, lineage, quality, and security are crucial, and data catalogs allow effective development but they are not enough. The real magic happens when data is used to make good business decisions.

A data-literate organization leverages the governance process to transform data into knowledge, overcoming market changes, perception bias, regulatory shifts, product versioning and privacy concerns.

ARTICLE PUBLISHED IN SEPTEMBER 2023

Happy Data Privacy Day, everyone! 🗝️

At Kaizen Gaming, we take data privacy very seriously. Every day is **"Data Privacy Day"** for us.

In today's digital world, our personal data is more valuable than ever. We're committed to protecting the data of our customers, employees, and partners. Everyone's data is safer when every company and individual handle personal data responsible, so here are a few simple things you can do to protect your privacy:

- * Be mindful of the information you share online.
- * Use strong passwords and two-factor authentication.
- * Be careful what apps you install on your devices.
- * Regularly review your privacy settings.

And if you work in a company that collects or handles personal data, it's essential to follow industry best practices. This includes:

- * Getting clear consent from individuals before collecting their data.
- * Using data only for its intended purpose.
- * Protecting data from unauthorized access, use, disclosure, alteration, or destruction.
- * Implementing appropriate security measures.

We believe that everyone deserves to have their data protected. Join us in making Data Privacy Day a day to celebrate and promote responsible data handling.

ARTICLE PUBLISHED IN JUNE 2024

Unleashing the Power of Data: A Mutant's Journey

Within my office, surrounded by data stacks that rival Stark Tower, I find myself on a new frontier – the information age. As Professor X, I have guided mutants through unimaginable challenges, but this uncharted territory demands telepathic abilities of a different kind.

Here, data flows like a mighty river, carrying valuable insights but also hidden dangers. How to navigate this digital sea without drowning in a tsunami of information? It's the question that haunts me as my telepathic powers scan the vastness of data in search of answers.

New Frontiers, New Challenges

Just as the X-Men have faced new enemies and challenges over the years, the information age presents its own dangers. Disinformation proliferates like a weed, and cybersecurity is a constant battle. We must be vigilant, adapting our skills and strategies to overcome these obstacles.

Xavier's Legacy in the Information Age

Xavier's legacy is based on the pursuit of knowledge and the use of power for good. In the information age, these principles are more important than ever. We must use our knowledge to combat disinformation and build a better future for all.

Move 1: Mapping the Unknown

To venture into the unknown, a map must be drawn. With my students – brilliant minds in data science and artificial intelligence – I define the objectives of our mission: to increase data reliability, accessibility, and value. More than just numbers, our goal is to transform data into knowledge, into insights that fuel innovation and illuminate the path to the future.

Uniting Forces for Success

Just as the X-Men unite their powers to face greater challenges, our team of data specialists works together to achieve our goals. Each team member contributes their unique skills and knowledge, creating a synergy that allows us to overcome any obstacle.

The Vision of a Data-Driven Future

I believe data can be a powerful force for good in the world. By using data responsibly and ethically, we can create a more just, sustainable, and prosperous future for all.

Move 2: Building the Bridge

The bridge connecting the present to the future is the data infrastructure. Like a mutant weaving their telepathic powers, I create a system to collect, store, and analyze data with the efficiency and precision of an X-Men strike.

Impenetrable Security, Absolute Trust

The security of our data infrastructure is a top priority. I implement rigorous cybersecurity measures to protect data against unauthorized access and malicious attacks. Trust in data security is fundamental to the success of our strategy.

Cutting-Edge Technology for a Bright Future

I invest in cutting-edge technologies to ensure that our data infrastructure is robust, scalable, and adaptable. I am always looking for new solutions to optimize data processing and analysis, driving innovation and growth.

Move 3: Cultivating the Collective Mind

For our data strategy to thrive, I cultivate a culture of data within the organization. I train our members like X-Men in development, teaching them to find, analyze, and interpret data, transforming it into weapons against inefficiency and uncertainty.

Continuous Learning and Evolution

Learning is an ongoing process in the information age. I encourage my collaborators to seek knowledge constantly, participating in training, workshops, and industry events. Through education and professional development, I cultivate a culture of continuous learning and improvement.

Data for All: Democratizing Knowledge

I believe that knowledge is power and that access to information should be democratized. Therefore, I strive to make data more accessible and understandable for all members of the organization, regardless of their position or area of expertise.

Move 4: The Journey Continues

The information age is still in its early stages, like a young mutant developing its powers. Our data strategy is a living organism, adapting and evolving as we unravel the secrets of the digital world.

With Optimism and Determination

I am optimistic about the future of data and its potential to make the world a better place. With the determination and unwavering resolve of Professor X, I will lead my team of data mutants towards the uncharted horizon of the data age, ready to face any challenge that comes our way.

ARTICLE PUBLISHED IN JUNE 2023

Crownpeak Customer Conference June 14, 2023

- We've been through difficult times with covid, now we're into a new wave of challenge with the economy. How is this impacting the priorities and investments of merchants?

- Pressure on efficiency

- Things changed. We need data

- Innovating and testing

- We hear how margin has become a key focus for merchants but equally price is now a sensitive driver for consumers. Where can merchants find margin improvement without impacting their pricing competitiveness?

- Services, identification of exact need.

-

- With ChatGPT I guess elevating AI into the mainstream. What practices are you seeing merchants embracing, the impact, and any lessons learnt?

- Changing mindsets

- Experimentation, innovation

- We're 3-4 months before we start to move into peak lock down of IT systems. What would you be shouting out about for merchants to be focusing on now to be ready to make peak success in 2023?

- Prepare infrastructure

- Test hypothesis

- Train

ARTICLE PUBLISHED IN JUNE 2023 (zero trust by design)

In the fast-paced world of artificial intelligence development, Trust emerges as the most valuable asset for both companies and individuals. If we want a team on the ground to carry out anything that came out from AI, people have to trust that they will get the best outcome from following suit. Think about these scenarios: A sales representative visiting a certain customer at a specific day, offering a particular product and charging a dynamic price for it. Or a technician installing equipment supported by AI interpreting a highly complex manual. Or an elderly customer who is trying to get someone to fix a botched purchase. In those cases if the sale does not go through or if the equipment isn't repaired properly, or if the customer quits the channel, trust will be lost.

We are humans and trust is what makes our societies work, it is the basis of our legal contracts, and it even grants our money worth. Without trust, pursuing business goals is futile. If it is shattered, rebuilding it is a costly endeavor. As more and more organizations jump onto the AI bandwagon, the question "can I trust AI" becomes ever more important. Answering this question in a practical perspective is a matter of continuously measuring how well AI responds to the problems we want it to tackle.

As large scale AI proof-of-concept projects are being deployed, we are getting both encouraging results that excite us and also anecdotal problems that challenge trust in machine learning and AI, like hallucination and poor accuracy.

Hallucination, an issue experienced in some AI models, occurs when the system generates false or misleading outputs that seem plausible at first glance. For instance, an image recognition algorithm might mistakenly identify a turtle as a rifle due to a specific combination of colors and patterns. Such errors, while rare, raise concerns about the reliability of AI predictions and the potential consequences they may have in critical decision-making scenarios.

Poor accuracy is inherent to the model that has been used. Despite significant advancements, AI models are not immune to inaccuracies. In certain cases, models may fail to provide accurate predictions due to various factors, such as insufficient training data or biases within the dataset. For example, an AI-powered diagnostic tool may incorrectly classify a benign tumor as malignant, potentially leading to unnecessary medical interventions or misdiagnoses.

Besides that the very nature of reality changes and AI might still be operating under circumstances that are no longer true. For instance, a dynamic pricing AI model may struggle to adapt to unforeseen economic events or shifts in market behavior, viral information, scandals, etc. This raises questions about the reliability of AI predictions when faced with unpredictable fluctuations and changing situations.

I've been contemplating the challenge of Trust for a while at CI&T as more and more of our teams started working to develop AI for our clients. I proactively decided to address it by getting inspiration from another area in the data landscape: Data Security. I came to adapt and utilize the Zero Trust perspective to all my developments. The results are amazing!

So, what exactly is this Zero Trust Approach?

The Zero Trust model acknowledges that trust is built when real world outcomes match predictions or interpretations. With this in mind, the developing team must establish a framework of continuous experimentation and verification of AI to ensure its reliability. Instead of blindly trusting the outputs of AI or language models like ChatGPT, the Zero Trust model assumes the presence of inaccuracies, errors, or false responses. It requires subjecting AI models to rigorous testing, using real-world data collected from the field.

Zero Trust also aligns with CI&T's core belief in experimentation, measurement, analysis and redeployment that propels every project we execute. The difference now is that the whole process is oriented towards AI development. By adopting the Zero Trust Approach, we shift from a mindset of hoping the predictions are accurate to understanding if it has met its purpose.

Zero Trust is embedded by design in the product development cycle with clear metrics, purpose and responsibilities. It has to be communicated to the entire group of stakeholders in the AI development, not only coders or data scientists. Its main tenants are:

Continuous Experiments: Establishing a culture of ongoing experimentation and validation for their AI models. Testing and verification against real-world scenarios will uncover potential weaknesses, biases, or inaccuracies. This allows for continuous improvement and enhances the overall trustworthiness of AI predictions and interpretations

Never Trust, Always Verify: Instead of assuming that everything generated by AI or machine learning models is accurate, the Zero Trust model promotes an attitude of skepticism. It encourages us to verify the outputs rigorously through meticulous testing and comparison. By keeping AI's results and assumptions in check we get ahead of hallucination and push the whole application to remain trustworthy and reliable.

Continuous Integration with Field Results: Truly understanding what success looks like in the field, seeking a more accurate representation of the challenges and complexities encountered in real life situations, not only to verify if the models worked but checking if the outputs were effective, useful and added value to whoever used the AI.

There are some other technical and non-technical issues regarding Zero Trust Approach, including ones related to compliance with proposed regulations. Implementing it the fullest even for us at CI&T who have a tradition of experimentation and customer centricity has been a challenging task. It involves a mindset shift from blind reliance to critical verification, enabling teams to identify and rectify weaknesses, enhance accuracy, and ensure that AI aligns with the desired outcomes.

I plan to discuss these issues in future articles.

For the time being I submit to you that the Zero Trust Approach is an effective way to foster the thing the AI systems will need the most.